

HW 11: Ch 9

- 1) Consider the signal $x[n] = \delta[n] - 2\delta[n-1] + 2\delta[n-2] - \delta[n-3]$.
 - (a) Compute and plot the 4-point DFT $X[k]$.
 - (b) Using the fewest number of zeros possible, zero pad $x[n]$ so that the DFT frequency resolution is at least $\Omega_0 = 0.1$. Compute and plot the corresponding DFT $X_{\text{pad}}[k]$.

- 2) The 8-point DFT of a real signal $x[n]$ is known over $0 \leq k \leq 4$ to be $X[k] = 1 \delta[k] + 2j\delta[k-2] + \delta[k-4]$.
 - (a) Determine $X[k]$ for $5 \leq k \leq 7$.
 - (b) Determine and plot the corresponding time-domain signal $x[n]$.

- 3) Consider a length-4 unit amplitude pulse $x[n] = u[n] - u[n-4]$.
 - (a) Compute and plot the 4-point DFT $X[k]$ of signal $x[n]$.
 - (b) Use the DFT interpolation formula to compute the DTFT $X(\Omega)$ from the DFT $X[k]$. Plot $X(\Omega)$.
 - (c) Sample $X(\Omega)$ over $0 \leq \Omega < 2\pi$ using 10 equally spaced points to create a signal $X_{10}[k]$. Compute and plot the IDFT of $X_{10}[k]$. Comment on your results.

- 4) This problem investigates zero padding applied in the frequency domain. Plot each N -point DFT as a function of frequency $f_k = k/N$.
 - a) In MATLAB, create a vector x that contains one period of the sinusoid $x[n] = \cos(\pi/2 n)$. Plot the result. How “sinusoidal” does the signal appear?
 - b) Using the `fft` command, compute the DFT $X[k]$ of the signal $x[n]$. Plot the magnitude of the DFT coefficients. Do they make sense?
 - c) Zero pad the DFT vector to a total length of 100 by inserting the appropriate number of zeros in the middle of the vector $X[k]$. Call this zero-padded DFT sequence Y . Why are zeros inserted in the middle rather than at the end? Take the inverse DFT of Y and plot the result. What similarities exist between the new signal y and the original signal x ? What are the differences between x and y ? What is the effect of zero padding in the frequency domain? How is this type of zero padding similar to zero padding in the time domain?
 - d) Derive a general modification to the procedure of zero padding in the frequency domain to ensure that the amplitude of the resulting time-domain signal is left unchanged.
 - e) Consider one period of a square wave described by the length-8 vector $[1, 1, 1, 1, -1, -1, -1, -1]$. Zero pad the DFT of this vector to a length of 100, and call the result S . Scale S according to (d), take the inverse DFT, and plot the result. Does the new time-domain signal $s[n]$ look like a square wave? Explain.